TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	58.9 ± 0.30
Correlation Stability	69.1 ± 0.30
Distribution Consistency	74.1 ± 1.04
Entropy Preservation	32.3 ± 1.10
Mean Z Score Drift	0.5 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	25.4 ± 5.39
Data Density	100.0 ± 0.00
Coverage Score	38.5 ± 8.57
Feature Accuracy	38.8 ± 0.40
Novelty Score	0.6 ± 0.04

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Dataset Audit: chronic_kidney_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	400
Columns	25
Numeric features	25
Categorical features	0
Missing cells (%)	10.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	51.483	44.952	-6.532	12.7%	Warn
bp	76.469	73.115	-3.354	4.4%	OK
sg	1.017	1.022	+0.005	0.5%	OK
al	1.017	0.000	-1.017	100.0%	Drift
su	0.450	0.000	-0.450	100.0%	Drift
rbc	0.190	0.000	-0.190	100.0%	Drift
pc	0.227	0.000	-0.227	100.0%	Drift
pcc	0.106	0.000	-0.106	100.0%	Drift
ba	0.056	0.000	-0.056	100.0%	Drift
bgr	148.037	106.031	-42.005	28.4%	Drift
bu	57.426	32.638	-24.787	43.2%	Drift
sc	3.072	0.966	-2.107	68.6%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	58.9 ± 0.30
Correlation Stability (*2)	69.1 ± 0.30
Distribution Consistency (*3)	74.1 ± 1.04
Entropy Preservation (*4)	32.3 ± 1.10
Mean Z Score Drift (*5)	0.5 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	25.4 ± 5.39

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	38.5 ± 8.57
Feature Accuracy (*10)	38.8 ± 0.40
Novelty Score (*11)	0.6 ± 0.04

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6024999999999999	N/A	0.5162500000000001	-0.08624999999999998	
F1	0.7507214882964476	N/A	0.4477184837772331	-0.3030030045192146	*

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9825000000000002	N/A	0.9887499999999999	0.006250000000000011	
F1	0.9848619694663338	N/A	0.9903499073254876	0.005487937859153791	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9974999999999999	N/A	0.9975000000000002	0	
F1	0.9978258300838945	N/A	0.9979146301726946	8.880088800089e-05	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6125	N/A	0.5525	-0.059999999999999984	
F1	0.5885198185573248	N/A	0.436411145964248	-0.15210867259307687	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7325	N/A	0.74	0.007500000000000018	
F1	0.7540259449990403	N/A	0.6948665199446415	-0.05915942505439877	

Model: DT

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.94625	N/A	0.9537500000000001	0.007500000000000018	
F1	0.9485222023533625	N/A	0.9578057020751828	0.009283499721820299	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9862500000000001	N/A	0.9862500000000001	0	
F1	0.9880875040761434	N/A	0.9882715024079728	0.00018399833182931592	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9525	N/A	0.9525	1.1102230246251566e-17	
F1	0.9587663012688215	N/A	0.9588769985542456	0.00011069728542402018	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9100000000000001	N/A	0.9112500000000001	0.0012500000000000289	
F1	0.92384446063128	N/A	0.9255684170474424	0.0017239564161624754	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.97875	N/A	0.9824999999999999	0.003749999999999977	
F1	0.9805471229310466	N/A	0.983846542508274	0.003299419577227491 5	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	48.0	80.0	1.02	1.0	0.0	nan	0.0	0.0	0.0	121.0
2	7.0	50.0	1.02	4.0	0.0	nan	0.0	0.0	0.0	nan
3	62.0	80.0	1.01	2.0	3.0	0.0	0.0	0.0	0.0	423.0
4	48.0	70.0	1.005	4.0	0.0	0.0	1.0	1.0	0.0	117.0
5	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
6	60.0	90.0	1.015	3.0	0.0	nan	nan	0.0	0.0	74.0
7	68.0	70.0	1.01	0.0	0.0	nan	0.0	0.0	0.0	100.0
8	24.0	nan	1.015	2.0	4.0	0.0	1.0	0.0	0.0	410.0
9	52.0	100.0	1.015	3.0	0.0	0.0	1.0	1.0	0.0	138.0
10	53.0	90.0	1.02	2.0	0.0	1.0	1.0	1.0	0.0	70.0
11	50.0	60.0	1.01	2.0	4.0	nan	1.0	1.0	0.0	490.0
12	63.0	70.0	1.01	3.0	0.0	1.0	1.0	1.0	0.0	380.0
13	68.0	70.0	1.015	3.0	1.0	nan	0.0	1.0	0.0	208.0
14	68.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	98.0
15	68.0	80.0	1.01	3.0	2.0	0.0	1.0	1.0	1.0	157.0
16	40.0	80.0	1.015	3.0	0.0	nan	0.0	0.0	0.0	76.0
17	47.0	70.0	1.015	2.0	0.0	nan	0.0	0.0	0.0	99.0
18	47.0	80.0	nan	nan	nan	nan	nan	0.0	0.0	114.0
19	60.0	100.0	1.025	0.0	3.0	nan	0.0	0.0	0.0	263.0
20	62.0	60.0	1.015	1.0	0.0	nan	1.0	1.0	0.0	100.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	32.2658961676807 65	80.0	1.02377649361280 1	0.0	0.0	0.0	0.0	0.0	0.0	89.0
2	52.6904838397174 9	78.7844651787986 8	1.02030388370530 05	0.0	0.0	0.0	0.0	0.0	0.0	117.756893035759 73
3	24.2569900032472 32	62.6606187520295 2	1.02433484531199 25	0.0	0.0	0.0	0.0	0.0	0.0	122.399092812804 43
4	25.0134645765022 16	70.8445538137518 4	1.025	0.0	0.0	0.0	0.0	0.0	0.0	135.354954024364 84
5	27.4026960298981 4	80.0	1.02133108664175 15	0.0	0.0	0.0	0.0	0.0	0.0	79.5391747314459 7
6	35.1801775620260 5	65.7557528466956 1	1.02287787642334 77	0.0	0.0	0.0	0.0	0.0	0.0	129.941946014626 13
7	35.0	80.0	1.005	3.0	0.0	1.0	0.0	0.0	0.0	nan
8	30.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	96.0
9	52.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	99.0
10	65.0	80.0	1.015	0.0	0.0	nan	0.0	0.0	0.0	115.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	41.2231009388580 25	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	97.8512660407613 2
2	49.1934188992350 7	63.2846929104816 95	1.02	0.0	0.0	0.0	0.0	0.0	0.0	127.357653544759 16
3	63.3674596841514 34	62.6471658947706 1	1.025	0.0	0.0	0.0	0.0	0.0	0.0	121.691042631536 73
4	52.2511884694023	73.1735340266931 9	1.01808231786262 22	0.0	0.0	0.0	0.0	0.0	0.0	95.8446911145817 8
5	49.3816481470818 5	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	111.222676764768 68
6	43.1558896264912	64.220551867544	1.02288972406622 8	0.0	0.0	0.0	0.0	0.0	0.0	94.7338344397368
7	64.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	113.0
8	42.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	98.0
9	12.0	60.0	1.015	3.0	0.0	1.0	1.0	1.0	0.0	nan
10	23.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	111.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	36.964136079568306	77.2401237086846	1.02	0.0	0.0	0.0	0.0	0.0	0.0	121.20790103304122
2	49.91167753463087	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	124.9750125399587
3	57.428899514717477	61.8774572815092 94	1.02	0.0	0.0	0.0	0.0	0.0	0.0	109.94266029116952
4	68.518062796592 29	74.0722511863691 6	1.02	0.0	0.0	0.0	0.0	0.0	0.0	84.3735604238539 7
5	69.111943752035 44	71.4019181542772 8	1.02070095907713 85	0.0	0.0	0.0	0.0	0.0	0.0	104.476547322300 9
6	75.209213673815 48	79.0523034184538 7	1.025	0.0	0.0	0.0	0.0	0.0	0.0	112.23200555600997
7	34.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	121.0
8	33.0	60.0	nan	nan	nan	0.0	0.0	0.0	0.0	130.0
9	60.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	124.0
10	57.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	132.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	38.605734049398 49	70.197132975300 75	1.02490143351234 96	0.0	0.0	0.0	0.0	0.0	0.0	123.66487394198872
2	55.0	80.0	1.01891450321718 02	0.0	0.0	0.0	0.0	0.0	0.0	102.31111672962675
3	44.961807971311 97	79.930559947839 95	1.02	0.0	0.0	0.0	0.0	0.0	0.0	116.90972793219194
4	54.803730395008 905	61.830224670825 91	1.02045755616770 65	0.0	0.0	0.0	0.0	0.0	0.0	103.53582026333928
5	39.181690510236 756	70.454226275591 89	1.02477288686220 4	0.0	0.0	0.0	0.0	0.0	0.0	123.54577372440811
6	46.401784678491 23	79.728083944768 74	1.02486404197238 42	0.0	0.0	0.0	0.0	0.0	0.0	123.02110220116747
7	44.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0
8	55.0	90.0	1.01	2.0	1.0	1.0	1.0	0.0	0.0	273.0
9	45.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	360.0
10	72.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	308.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	25.561115385553954	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	119.75307692099122
2	65.12659530279936	68.54432550466561	1.0242721627523328	0.0	0.0	0.0	0.0	0.0	0.0	110.93032113740286
3	34.02850944353102	70.2850944353102	1.01873775868559 15	0.0	0.0	0.0	0.0	0.0	0.0	116.62937723409674
4	45.20036399049032	80.0	1.0226382591499479	0.0	0.0	0.0	0.0	0.0	0.0	71.889392680041 72
5	29.947574839213864	66.10485032157227	1.0181946266098818	0.0	0.0	0.0	0.0	0.0	0.0	120.53748200156298
6	64.93235926629704	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	114.45333499784681
7	41.0	80.0	1.015	1.0	4.0	1.0	0.0	0.0	0.0	210.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	87.0
9	26.0	70.0	1.015	0.0	4.0	nan	0.0	0.0	0.0	250.0
10	50.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	137.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	24.3183990998650 4	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	104.72068610005783
2	29.1913150980413 47	74.9043424509793 3	1.02254782877451 02	0.0	0.0	0.0	0.0	0.0	0.0	115.662617833335 15
3	61.5032552017837 94	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	130.344184005945 98
4	65.0	67.4565061724261 2	1.02372825308621 3	0.0	0.0	0.0	0.0	0.0	0.0	91.1043851861773 2
5	24.7599974138447	74.2400025861553	1.02	0.0	0.0	0.0	0.0	0.0	0.0	118.519994827689 4
6	29.8511052802819 35	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	88.1066316816916 1
7	8.0	60.0	1.025	3.0	0.0	0.0	0.0	0.0	0.0	78.0
8	60.0	70.0	1.01	1.0	0.0	nan	0.0	0.0	0.0	109.0
9	nan	70.0	1.01	3.0	0.0	0.0	0.0	0.0	0.0	110.0
10	62.0	90.0	1.02	2.0	1.0	nan	0.0	0.0	0.0	169.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	55.4998223949441 4	78.3339253501862	1.025	0.0	0.0	0.0	0.0	0.0	0.0	123.002486470782 05
2	28.1517605472696 67	62.1011736981797 8	1.02052529342454 5	0.0	0.0	0.0	0.0	0.0	0.0	120.155645533998 88
3	54.9401454564356 94	80.0	1.02228467132939 83	0.0	0.0	0.0	0.0	0.0	0.0	101.894892301294 58
4	53.7669875983841 4	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	126.236891318868 9
5	31.5175375275523 24	80.0	1.02216082082414 93	0.0	0.0	0.0	0.0	0.0	0.0	104.185969977958 14
6	28.1267580090261 05	69.9332852584073 1	1.02	0.0	0.0	0.0	0.0	0.0	0.0	130.853227568496 1
7	nan	70.0	1.02	0.0	0.0	nan	nan	0.0	0.0	219.0
8	29.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	127.0
9	50.0	90.0	1.015	1.0	0.0	1.0	1.0	0.0	0.0	nan
10	65.0	70.0	1.02	1.0	0.0	1.0	1.0	0.0	0.0	139.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	39.7314352965530 5	77.3143529655305 5	1.02134282351723 48	0.0	0.0	0.0	0.0	0.0	0.0	120.342823517234 73
2	46.4819754745542 54	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	96.7760840872425 5
3	53.9741021721821 4	69.9136739072738	1.02004316304636 3	0.0	0.0	0.0	0.0	0.0	0.0	76.1985500132702 5
4	51.4482601432203 1	60.6403716331718 7	1.02	0.0	0.0	0.0	0.0	0.0	0.0	127.270996590435 96
5	55.4263258019224 4	76.1929713311972 8	1.01952579116581 58	0.0	0.0	0.0	0.0	0.0	0.0	126.477188532478 92
6	36.0928761994304 9	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	104.152130654458 03
7	53.0	100.0	1.01	1.0	3.0	1.0	0.0	0.0	0.0	213.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	nan
9	45.0	80.0	1.02	3.0	0.0	0.0	1.0	0.0	0.0	425.0
10	33.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	89.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	61.3133700331013	70.0	1.02109447502758	0.0	0.0	0.0	0.0	0.0	0.0	115.592154939314
	86		45							13
2	34.7121311292379	77.1213112923791	1.01960102384578	0.0	0.0	0.0	0.0	0.0	0.0	107.742295319907
	2	5	6							11
3	26.5088599639971	79.5808722322230	1.025	0.0	0.0	0.0	0.0	0.0	0.0	120.958087223222
	1	3								31
4	40.5774872661218	66.2642735867827	1.01913179366077	0.0	0.0	0.0	0.0	0.0	0.0	93.8189231140522
	64	2	93							3
5	36.1091742305371	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	103.112660615570
	8									26
6	47.4998578532577	62.9167851222852	1.02145839256114	0.0	0.0	0.0	0.0	0.0	0.0	127.541607438857
	44	1	27							4
7	46.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	102.0
8	22.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	97.0
9	65.0	80.0	1.01	3.0	3.0	nan	0.0	0.0	0.0	294.0
10	23.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	72.7528183062885	68.7515657257158	1.02	0.0	0.0	0.0	0.0	0.0	0.0	106.750313145143
	3	5								17
2	57.4299166588844	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	105.114644442369
	9									2
3	36.5347567268530	67.2608408076145	1.02363042040380	0.0	0.0	0.0	0.0	0.0	0.0	127.834822869339
	54		71							7
4	68.0991150574854	65.1651917624756	1.02	0.0	0.0	0.0	0.0	0.0	0.0	94.5374630650681
		7								1
5	39.4776419288981	70.5970524111226	1.02470147379443	0.0	0.0	0.0	0.0	0.0	0.0	122.268548007744
	4	6	87							27
6	51.5925188512330	75.8249828625154	1.01835200662786	0.0	0.0	0.0	0.0	0.0	0.0	96.7575154237361
	6		48							4
7	11.0	80.0	1.01	3.0	0.0	nan	0.0	0.0	0.0	nan
8	61.0	90.0	1.015	0.0	2.0	nan	0.0	0.0	0.0	nan
9	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
10	70.0	60.0	nan	nan	nan	nan	nan	0.0	0.0	120.0